

Algebra 1 – Lab 2: Balance → Equations

Name: _____ Period: _____

Module 2 — Solving equations Materials: algebra tiles · pencil. The mat IS the equation. Every move you make on it is one of our five rules.

 = x  = 1  = -x  = -1  = a tile you ADDED *X = zero pair*

THE FIVE RULES ON THE MAT: 1 box the variable = **ONE bar alone** · 2 stay equal = **same move BOTH sides** · 3 opposites make ZERO = **zero pairs cancel** · 4 check by substitution = **build it back in** · 5 side by side, divide = **split into equal groups**

Part 1 — One move. (Rules 1, 3, 5 — then 4) Build both sides. BOX THE BAR: your goal is ONE bar alone (Rule 1). Add tiles to BOTH sides (Rule 2) so opposites make ZERO (Rule 3) — or split both sides into equal groups (Rule 5). Then build the answer back in (Rule 4).

a. $x + 3 = 7$ <i>balance mat — build here</i> Move 1: _____ x = _____ check: _____ ★ Think deeper: Why can't you just take the plain squares off the left side and be done? _____ _____	b. $x - 2 = 3$ <i>balance mat — build here</i> Move 1: _____ x = _____ check: _____ ★ Think deeper: In (a) you added flipped squares; in (b) plain ones. What decides which kind to add? _____ _____	c. $3x = 9$ <i>balance mat — build here</i> Move 1: _____ x = _____ check: _____ ★ Think deeper: Could you split $3x = 9$ into 2 groups instead? What would one group tell you? _____ _____
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Part 2 — Two moves. (Rule 3, then Rule 5) Squares first (Rule 3), then split (Rule 5). Write BOTH moves. The order matters: try splitting first and see what happens to the loose squares.

a. $2x + 1 = 7$ <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Rex split FIRST: 2 groups of (1 bar + $\frac{1}{2}$ square)... Why is that a bad first move here? _____ _____	b. $3x - 2 = 4$ <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Write the same two moves as ALGEBRA steps — one line each, no tiles. _____ _____	c. $2x + 5 = 1$ <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Rule 4 on the mat: BUILD your answer — replace each bar with its value. Does it balance? _____ _____
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Part 3 — Bars on both sides. (Rule 2 does the work) Same rules, one more move: take the SAME number of bars off BOTH sides first — that is Rule 2, and it is the only thing keeping the mat level. Now it looks like Part 2.

a. $3x + 1 = x + 5$ <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Does it matter whether you remove the bars or the squares first? Try the other order. _____ _____	b. $4x - 2 = 2x + 4$ <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Which side should keep the bars — the one with more, or fewer? Why? _____ _____	c. $x + 6 = 3x + 2$ <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Marcus ended with $4 = 2x$ and wrote $x = 2$. Ben ended with $2x = 4$ and wrote $x = 2$. Who's right? _____
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Part 4 — Team Challenge, then the No-Tiles Check. (all five) Work with your group on a-c. Then put the tiles AWAY for the check on the right — the rules do not need tiles.

a. $2(x + 2) = 8$ (Hint: rows, like Lab 1.) <i>balance mat — build here</i> Move 1: _____ Move 2: _____ x = _____ check: _____ ★ Think deeper: Rex split into 2 groups FIRST: $x + 2 = 4$. Same answer. Why does that work here — and when would it not be easier? _____ _____	b. $2x + 1 = 2x + 5$ Build it, make your moves, and STOP when something strange happens. What does the mat say? <i>balance mat — build here</i> _____ c. Design one: write an equation with bars on BOTH sides whose solution is $x = -2$. Trade with a partner and solve theirs with tiles. _____ _____	No-Tiles Check. Pencil only. Write each move as an algebra step, NAME the rule you used, then finish with Rule 4. 1. $3x + 5 = 20$ x = _____ 2. $4x - 7 = 2x + 9$ x = _____ 3. $2(x - 4) = 6$ x = _____ 4. $5x + 8 = 3x + 2$ x = _____ ★ Think deeper: Which of the four would need the MOST tiles? Which is the hardest? Are they the same one? _____ _____
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